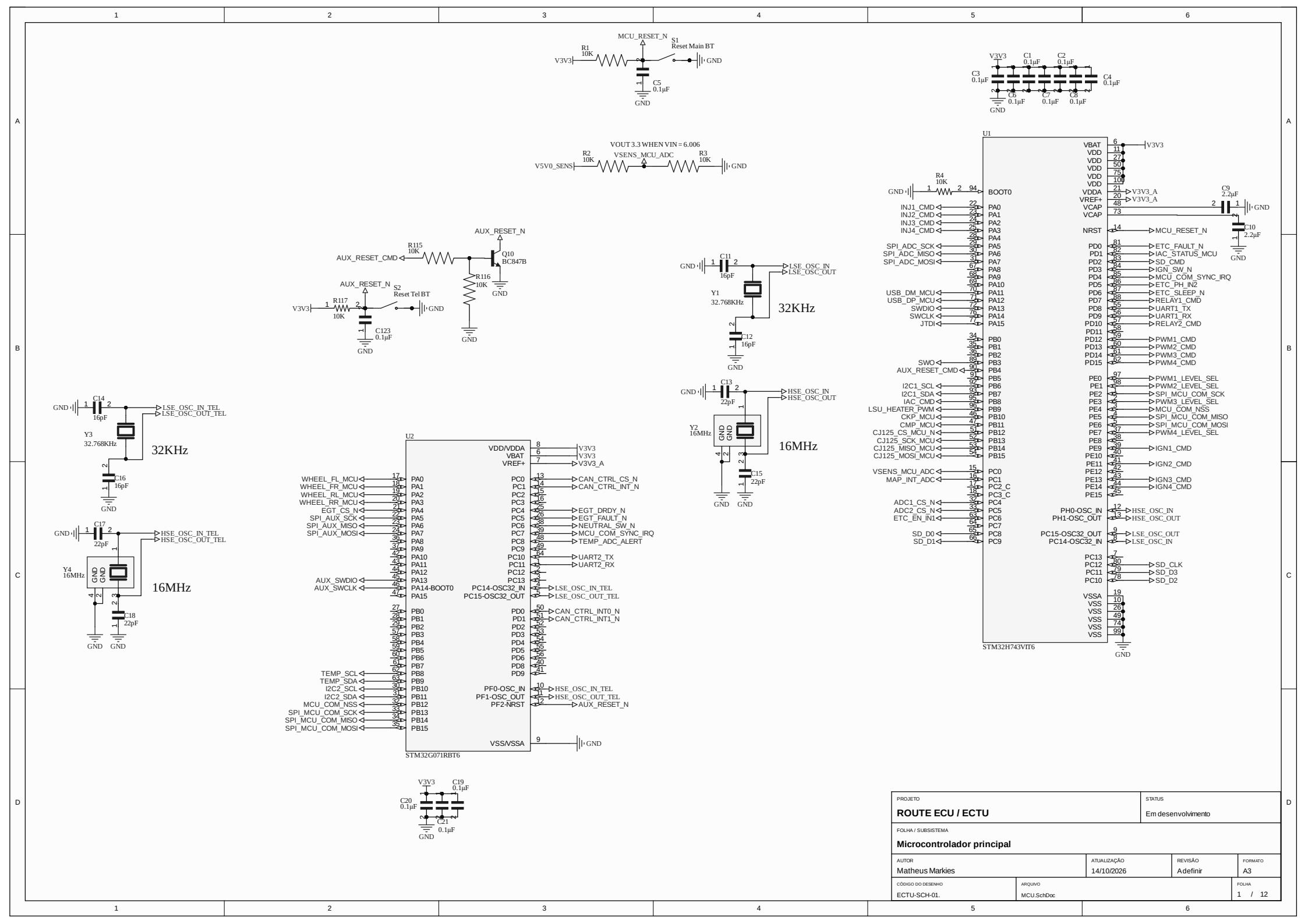
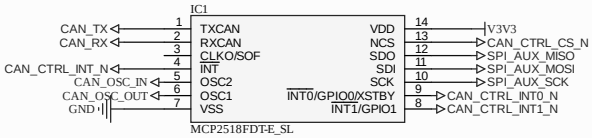
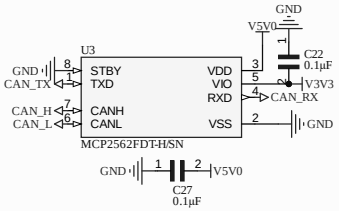
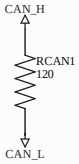
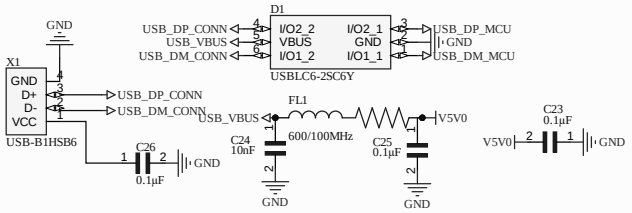

MCU_RESET_N
S1
Reset Main BT
R1
10K
V3V3
GND
C5
0.1µF
V3V3
C1
0.1µF
C2
0.1µF
C3
0.1µF
C4
0.1µF
C6
0.1µF
C7
0.1µF
C8
0.1µF
VOUT 3.3 WHEN VIN = 6.006
R2
10K
VSENS_MCU_ADC
R3
10K
V5V0_SENS
AUX_RESET_N
R115
10K
AUX_RESET_CMD
Q10
BC847B
R116
10K
S2
Reset Tel BT
R117
10K
C123
0.1µF
C11
16pF
LSE_OSC_IN
LSE_OSC_OUT
Y1
32.768KHz
32KHz
C12
16pF
C13
22pF
HSE_OSC_IN
HSE_OSC_OUT
Y2
16MHz
16MHz
C15
22pF
C14
16pF
LSE_OSC_IN_TEL
LSE_OSC_OUT_TEL
Y3
32.768KHz
32KHz
C16
16pF
C17
22pF
HSE_OSC_IN_TEL
HSE_OSC_OUT_TEL
Y4
16MHz
16MHz
C18
22pF
U2
VDD/VDDA
VBAT
VREF+
PA0
PA1
PA2
PA3
PA4
PA5
PA6
PA7
PA8
PA9
PA10
PA11
PA12
PA13
PA14-BOOT0
PA15
PB0
PB1
PB2
PB3
PB4
PB5
PB6
PB7
PB8
PB9
PB10
PB11
PB12
PB13
PB14
PB15
PC0
PC1
PC2
PC3
PC4
PC5
PC6
PC7
PC8
PC9
PC10
PC11
PC12
PC13
PC14-OSC32_IN
PC15-OSC32_OUT
PD0
PD1
PD2
PD3
PD4
PD5
PD6
PD8
PD9
PF0-OSC_IN
PF1-OSC_OUT
PF2-NRST
VSS/VSSA
WHEEL_FL_MCU
WHEEL_FR_MCU
WHEEL_RL_MCU
WHEEL_RR_MCU
EGT_CS_N
SPI_AUX_SCK
SPI_AUX_MISO
SPI_AUX_MOSI
AUX_SWDIO
AUX_SWCLK
TEMP_SCL
TEMP_SDA
I2C2_SCL
I2C2_SDA
MCU_COM_NSS
SPI_MCU_COM_SCK
SPI_MCU_COM_MISO
SPI_MCU_COM_MOSI
CAN_CTRL_CS_N
CAN_CTRL_INT_N
EGT_DRDY_N
EGT_FAULT_N
NEUTRAL_SW_N
MCU_COM_SYNC_IRQ
TEMP_ADC_ALERT
UART2_TX
UART2_RX
LSE_OSC_IN_TEL
LSE_OSC_OUT_TEL
CAN_CTRL_INT0_N
CAN_CTRL_INT1_N
HSE_OSC_IN_TEL
HSE_OSC_OUT_TEL
AUX_RESET_N
V3V3_A
GND
STM32G071RBT6
V3V3
C19
0.1µF
C20
0.1µF
C21
0.1µF
U1
VBAT
VDD
VDDA
VREF+
VCAP
NRST
BOOT0
PA0
PA1
PA2
PA3
PA4
PA5
PA6
PA7
PA8
PA9
PA10
PA11
PA12
PA13
PA14
PA15
PB0
PB1
PB2
PB3
PB4
PB5
PB6
PB7
PB8
PB9
PB10
PB11
PB12
PB13
PB14
PB15
PC0
PC1
PC2_C
PC3_C
PC4
PC5
PC6
PC7
PC8
PC9
PD0
PD1
PD2
PD3
PD4
PD5
PD6
PD7
PD8
PD9
PD10
PD11
PD12
PD13
PD14
PD15
PE0
PE1
PE2
PE3
PE4
PE5
PE6
PE7
PE8
PE9
PE10
PE11
PE12
PE13
PE14
PE15
PH0-OSC_IN
PH1-OSC_OUT
PC15-OSC32_OUT
PC14-OSC32_IN
PC13
PC12
PC11
PC10
VSSA
VSS
R4
10K
C9
2.2µF
C10
2.2µF
MCU_RESET_N
ETC_FAULT_N
IAC_STATUS_MCU
SD_CMD
IGN_SW_N
MCU_COM_SYNC_IRQ
ETC_PH_IN2
ETC_SLEEP_N
RELAY1_CMD
UART1_TX
UART1_RX
RELAY2_CMD
PWM1_CMD
PWM2_CMD
PWM3_CMD
PWM4_CMD
PWM1_LEVEL_SEL
PWM2_LEVEL_SEL
SPI_MCU_COM_SCK
PWM3_LEVEL_SEL
MCU_COM_NSS
SPI_MCU_COM_MISO
SPI_MCU_COM_MOSI
PWM4_LEVEL_SEL
IGN1_CMD
IGN2_CMD
IGN3_CMD
IGN4_CMD
HSE_OSC_IN
HSE_OSC_OUT
LSE_OSC_OUT
LSE_OSC_IN
SD_CLK
SD_D3
SD_D2
INJ1_CMD
INJ2_CMD
INJ3_CMD
INJ4_CMD
SPI_ADC_SCK
SPI_ADC_MISO
SPI_ADC_MOSI
USB_DM_MCU
USB_DP_MCU
SWDIO
SWCLK
JTDI
SWO
AUX_RESET_CMD
I2C1_SCL
I2C1_SDA
IAC_CMD
LSU_HEATER_PWM
CKP_MCU
CMP_MCU
CJ125_CS_MCU_N
CJ125_SCK_MCU
CJ125_MISO_MCU
CJ125_MOSI_MCU
VSENS_MCU_ADC
MAP_INT_ADC
ADC1_CS_N
ADC2_CS_N
ETC_EN_IN1
SD_D0
SD_D1
SD_CLK
SD_D3
SD_D2
STM32H743VIT6
PROJETO
ROUTE ECU / ECTU
STATUS
Em desenvolvimento
FOLHA / SUBSISTEMA
Microcontrolador principal
AUTOR
Matheus Markies
ATUALIZAÇÃO
14/10/2026
REVISÃO
A definir
FORMATO
A3
CÓDIGO DO DESENHO
ECTU-SCH-01.
ARQUIVO
MCU.SchDoc
FOLHA
1 / 12

CAN

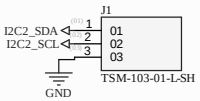


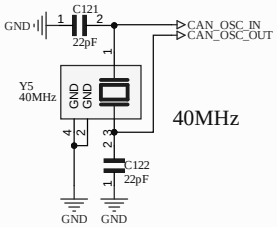




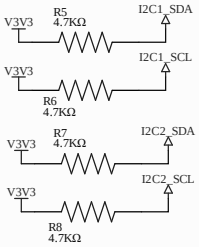


I2C

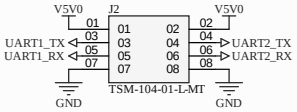


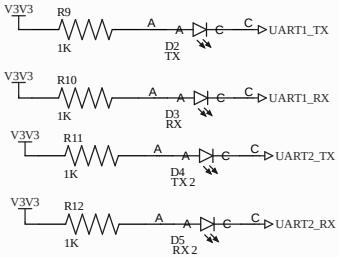


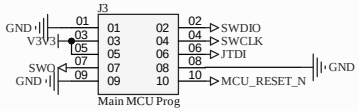
40MHz

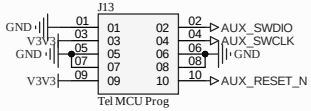


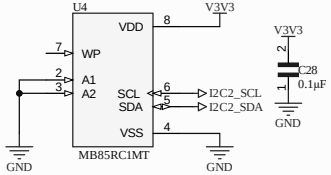
UART

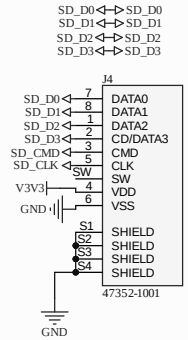




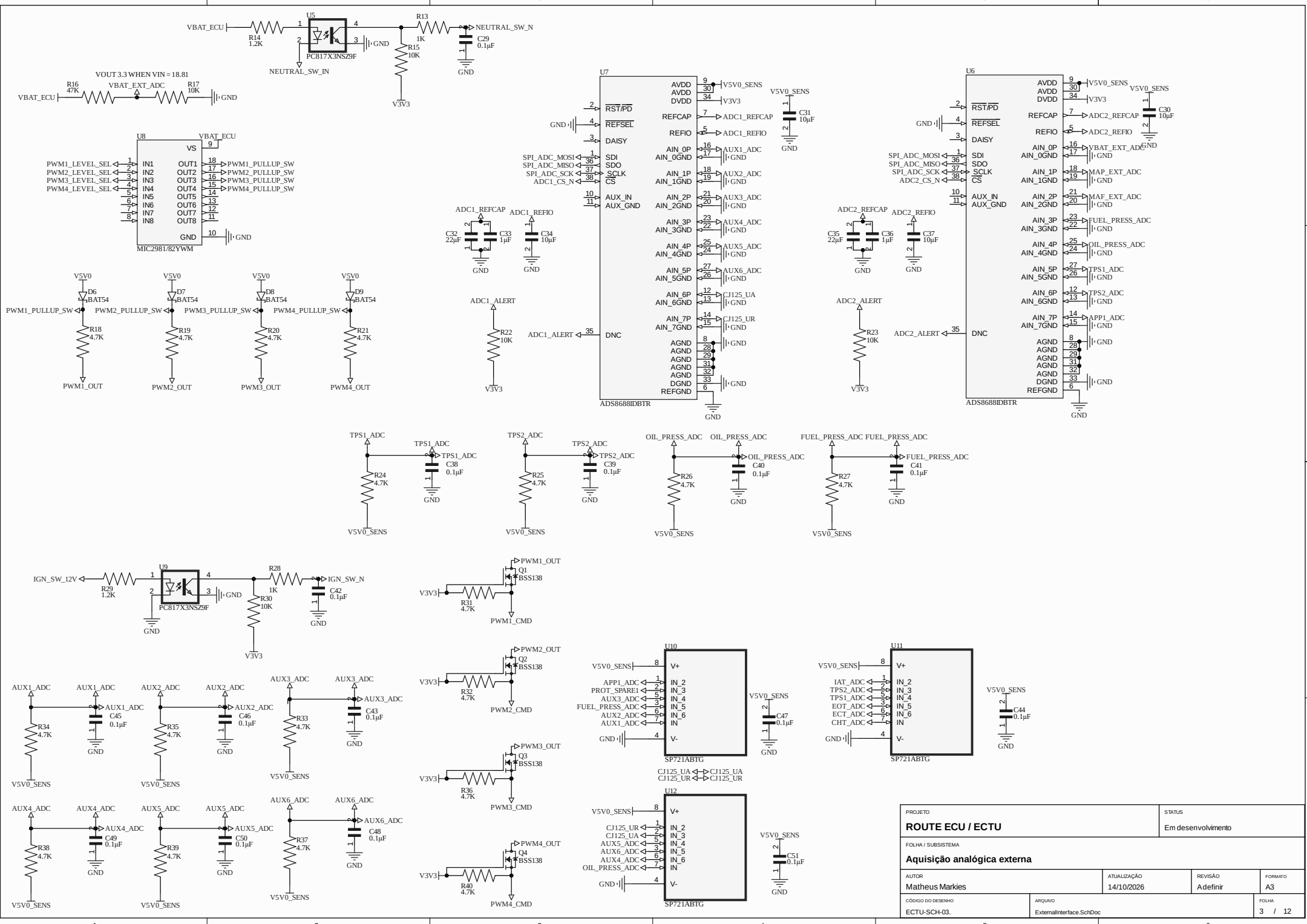


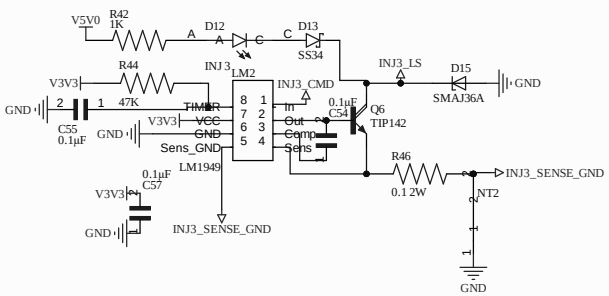


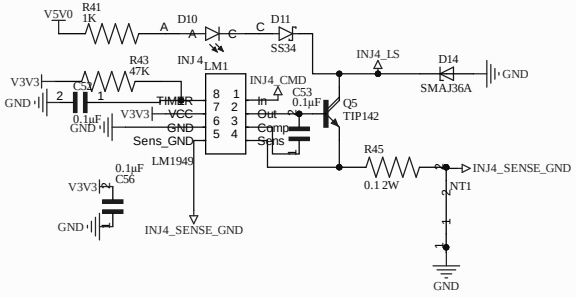


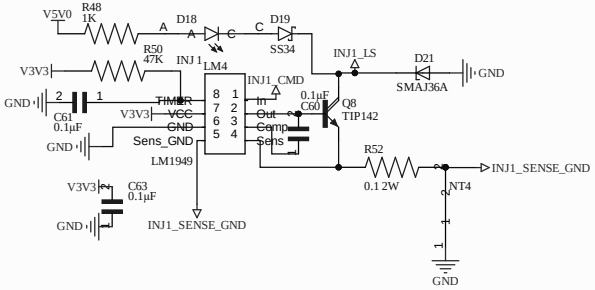


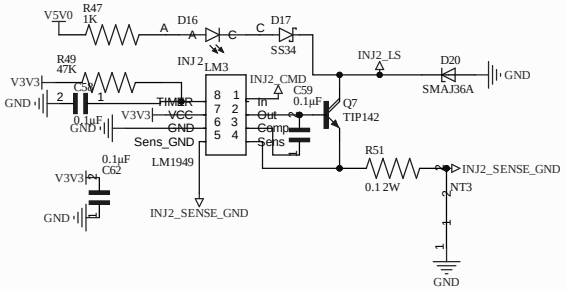
PROJETO: **ROUTE ECU / ECTU**		STATUS: Em desenvolvimento	
FOLHA / SUBSISTEMA: **Interfaces de comunicação**			
AUTOR: Matheus Markies	ATUALIZAÇÃO: 14/10/2026	REVISÃO: Adefinir	FORMATO: A3
CÓDIGO DO DESENHO: ECTU-SCH-02.	ARQUIVO: CommunicationInterfaces.SchDoc		FOLHA: 2 / 12

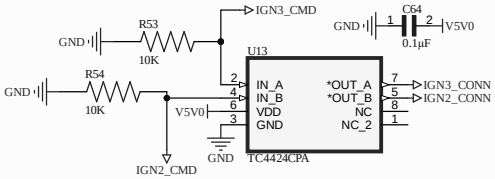

VBAT_ECU
R14 1.2K
U5
PC817X3NSZ9F
NEUTRAL_SW_IN
R13 1K
R15 10K
C29 0.1µF
NEUTRAL_SW_N
GND
V3V3
VOUT 3.3 WHEN VIN = 18.81
R16 47K
VBAT_EXT_ADC
R17 10K
U8
MIC2981/82YWM
PWM1_LEVEL_SEL
PWM2_LEVEL_SEL
PWM3_LEVEL_SEL
PWM4_LEVEL_SEL
PWM1_PULLUP_SW
PWM2_PULLUP_SW
PWM3_PULLUP_SW
PWM4_PULLUP_SW
U7
ADS8688IDBTR
U6
ADS8688IDBTR
V5V0_SENS
V3V3
ADC1_REFCAP
ADC1_REFIO
ADC2_REFCAP
ADC2_REFIO
SPI_ADC_MOSI
SPI_ADC_MISO
SPI_ADC_SCK
ADC1_CS_N
ADC2_CS_N
ADC1_ALERT
ADC2_ALERT
D6 BAT54
D7 BAT54
D8 BAT54
D9 BAT54
R18 4.7K
R19 4.7K
R20 4.7K
R21 4.7K
R22 10K
R23 10K
PWM1_OUT
PWM2_OUT
PWM3_OUT
PWM4_OUT
TPS1_ADC
TPS2_ADC
OIL_PRESS_ADC
FUEL_PRESS_ADC
R24 4.7K
R25 4.7K
R26 4.7K
R27 4.7K
C38 0.1µF
C39 0.1µF
C40 0.1µF
C41 0.1µF
IGN_SW_12V
U9
PC817X3NSZ9F
R29 1.2K
R28 1K
R30 10K
C42 0.1µF
IGN_SW_N
Q1 BSS138
Q2 BSS138
Q3 BSS138
Q4 BSS138
R31 4.7K
R32 4.7K
R36 4.7K
R40 4.7K
PWM1_CMD
PWM2_CMD
PWM3_CMD
PWM4_CMD
AUX1_ADC
AUX2_ADC
AUX3_ADC
AUX4_ADC
AUX5_ADC
AUX6_ADC
R34 4.7K
R35 4.7K
R33 4.7K
R38 4.7K
R39 4.7K
R37 4.7K
C45 0.1µF
C46 0.1µF
C43 0.1µF
C49 0.1µF
C50 0.1µF
C48 0.1µF
U10
U11
U12
SP721ABTG
CJ125_UA
CJ125_UR
APP1_ADC
PROT_SPARE1
FUEL_PRESS_ADC
IAT_ADC
EOT_ADC
ECT_ADC
CHT_ADC
C47 0.1µF
C44 0.1µF
C51 0.1µF
PROJETO
ROUTE ECU / ECTU
STATUS
Em desenvolvimento
FOLHA / SUBSISTEMA
Aquisição analógica externa
AUTOR
Matheus Markies
ATUALIZAÇÃO
14/10/2026
REVISÃO
Adefinir
FORMATO
A3
CÓDIGO DO DESENHO
ECTU-SCH-03.
ARQUIVO
ExternalInterface.SchDoc
FOLHA
3 / 12

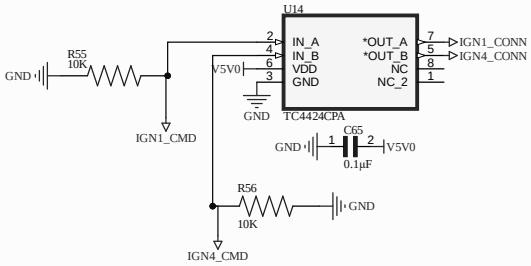




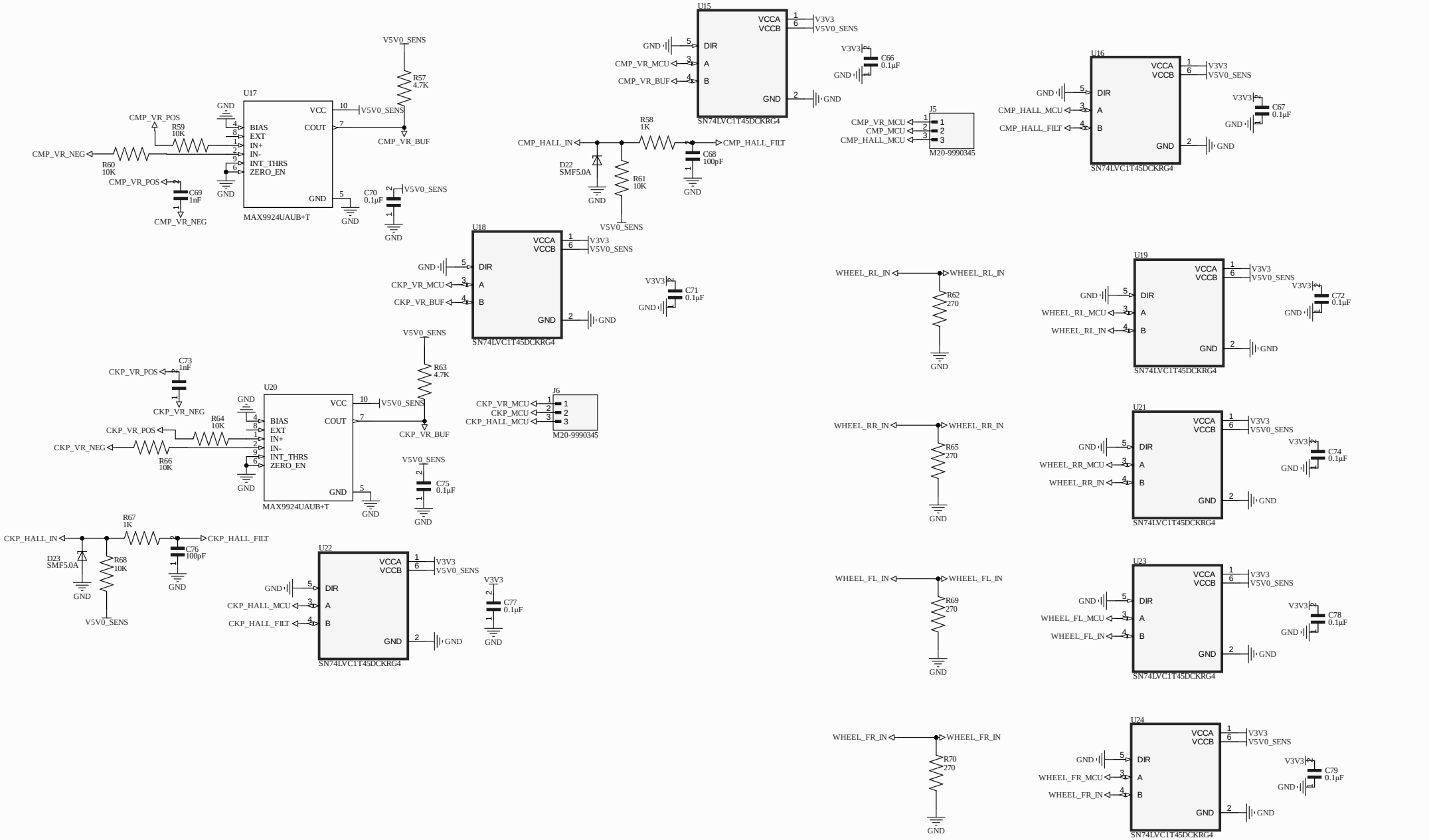






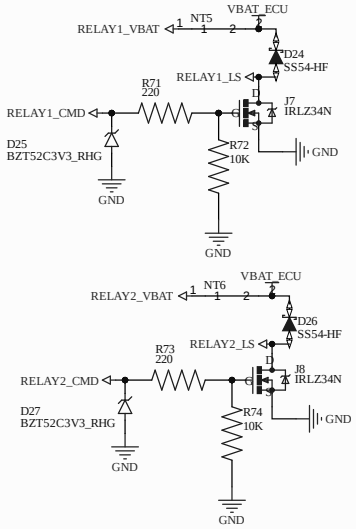


PROJETO: **ROUTE ECU / ECTU**			STATUS: Em desenvolvimento
FOLHA / SUBSISTEMA: **Injeção e ignição**			
AUTOR: Matheus Markies	ATUALIZAÇÃO: 14/10/2026	REVISÃO: A definir	FORMATO: A3
CÓDIGO DO DESENHO: ECTU-SCH-04.	ARQUIVO: SparkAndFuel.SchDoc		FOLHA: 4 / 12

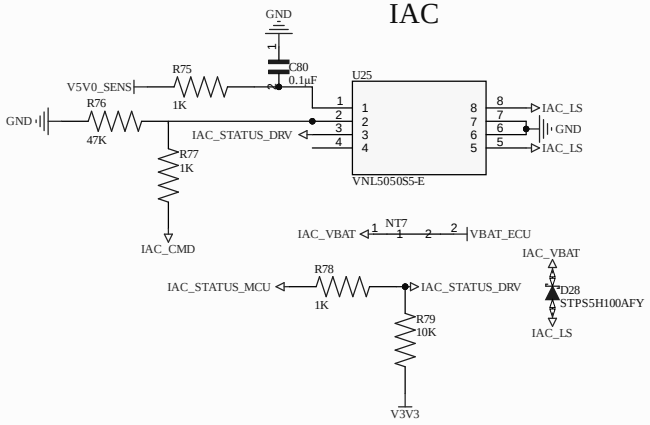


PROJETO	**ROUTE ECU / ECTU**
STATUS	Em desenvolvimento
FOLHA / SUBSISTEMA	**Rotação e fase (CKP / CMP)**
AUTOR	Matheus Markies
ATUALIZAÇÃO	14/10/2026
REVISÃO	A definir
FORMATO	A3
CÓDIGO DO DESENHO	ECTU-SCH-05.
ARQUIVO	CrankAndCam.SchDoc
FOLHA	5 / 12

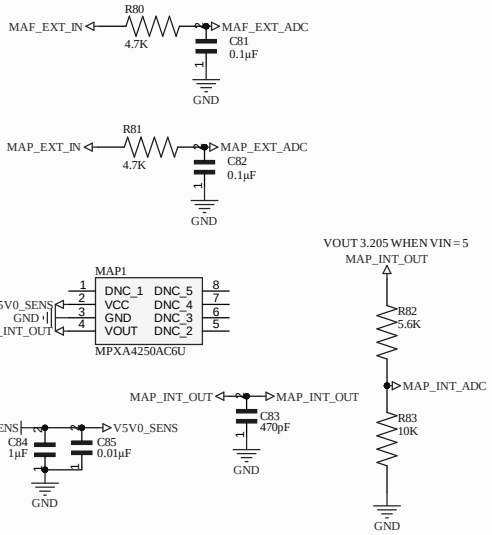
Fuel And Starter



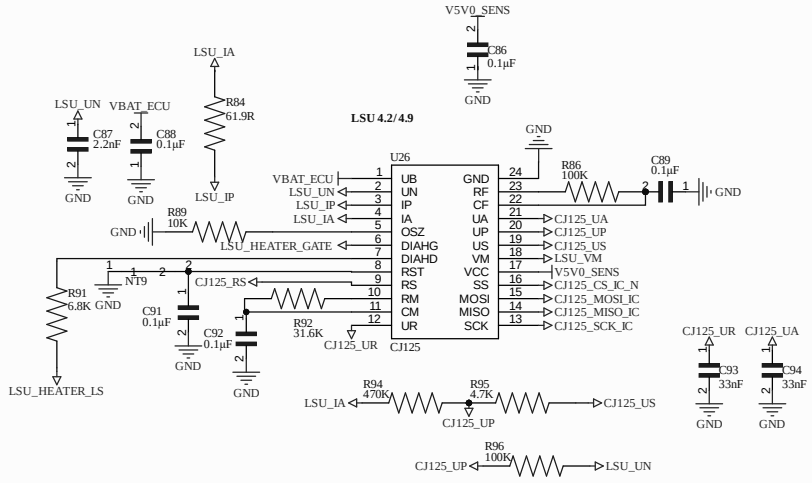
IAC

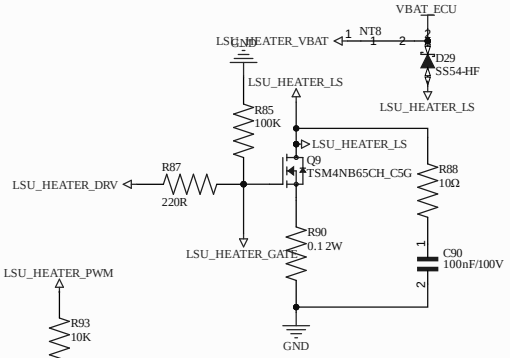


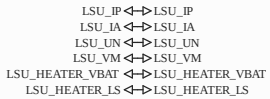
PROJETO: **ROUTE ECU / ECTU**	STATUS: Em desenvolvimento
FOLHA / SUBSISTEMA: **Acionamentos auxiliares**	
AUTOR: Matheus Markies	ATUALIZAÇÃO: 14/10/2026
REVISÃO: A definir	FORMATO: A3
CÓDIGO DO DESENHO: ECTU-SCH-06.	ARQUIVO: LowSideSwitches.SchDoc
FOLHA: 6 / 12	

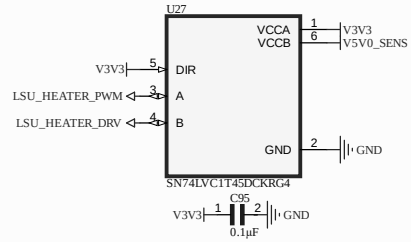


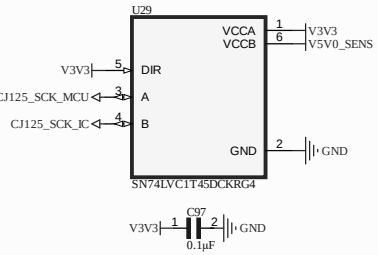
PROJETO: **ROUTE ECU / ECTU**			STATUS: Em desenvolvimento
FOLHA / SUBSISTEMA: **Pressão e fluxo de ar**			
AUTOR: Matheus Markies	ATUALIZAÇÃO: 14/10/2026	REVISÃO: A definir	FORMATO: A3
CÓDIGO DO DESENHO: ECTU-SCH-07.	ARQUIVO: PressureAirflow.SchDoc		FOLHA: 7 / 12

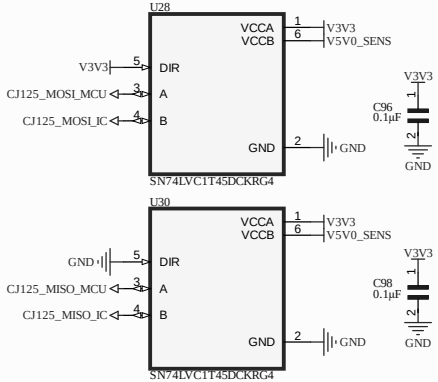


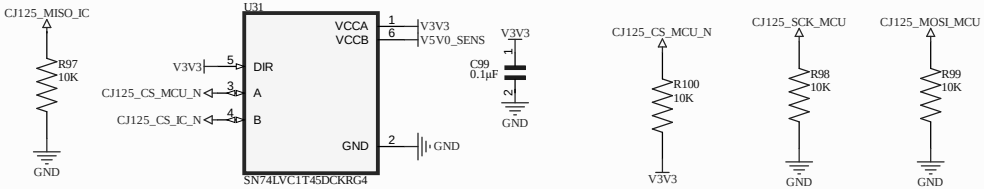




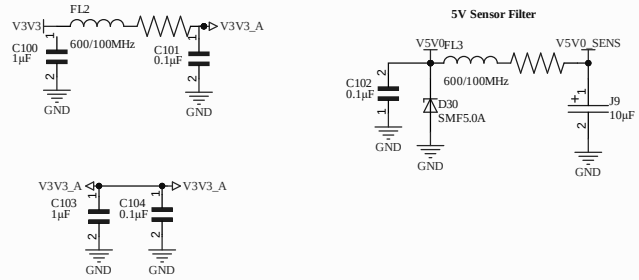


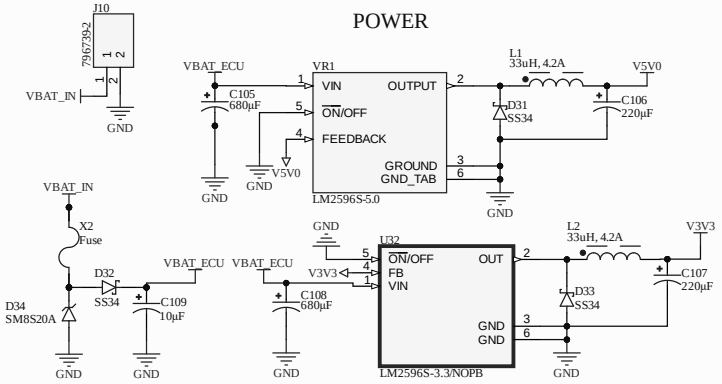


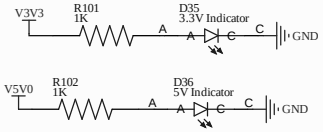




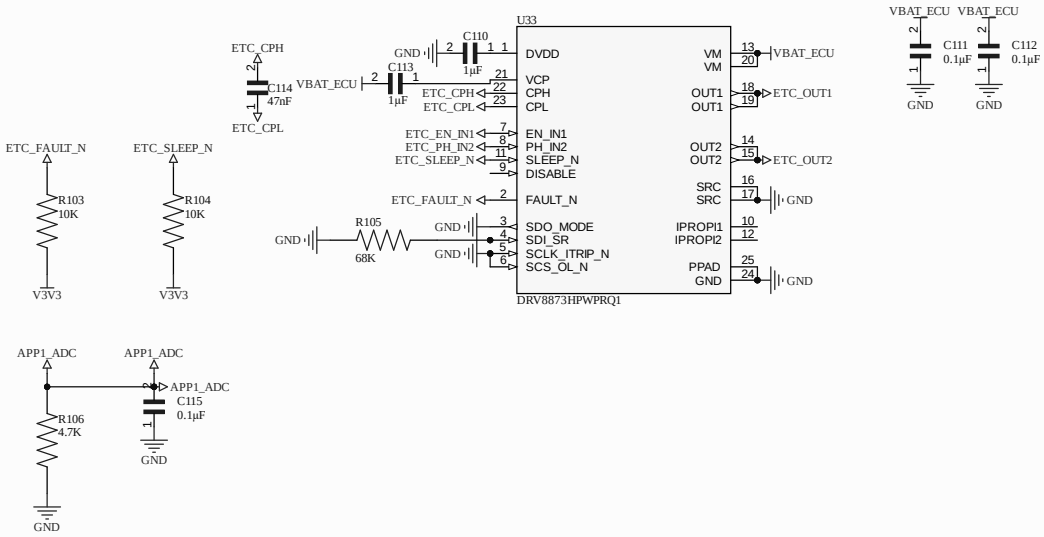
PROJETO: **ROUTE ECU / ECTU**			STATUS: Em desenvolvimento
FOLHA / SUBSISTEMA: **Controle da sonda lambda**			
AUTOR: Matheus Markies	ATUALIZAÇÃO: 14/10/2026	REVISÃO: Adefinir	FORMATO: A3
CÓDIGO DO DESENHO: ECTU-SCH-08.	ARQUIVO: LambdaWidebandController.SchDoc		FOLHA: 8 / 12







PROJETO: **ROUTE ECU / ECTU**			STATUS: Em desenvolvimento
FOLHA / SUBSISTEMA: **Alimentação e proteções**			
AUTOR: Matheus Markies	ATUALIZAÇÃO: 14/10/2026	REVISÃO: Adefinir	FORMATO: A3
CÓDIGO DO DESENHO: ECTU-SCH-09.	ARQUIVO: PowerManagement.SchDoc		FOLHA: 9 / 12

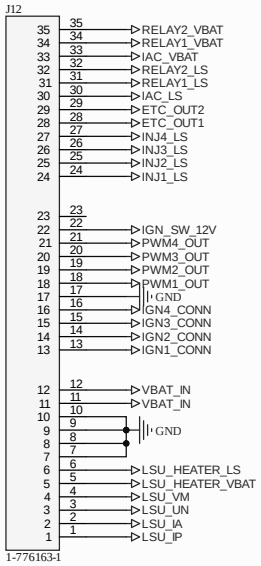


PROJETO	STATUS
ROUTE ECU / ECTU	Em desenvolvimento
FOLHA / SUBSISTEMA	
Controle eletrônico da borboleta	

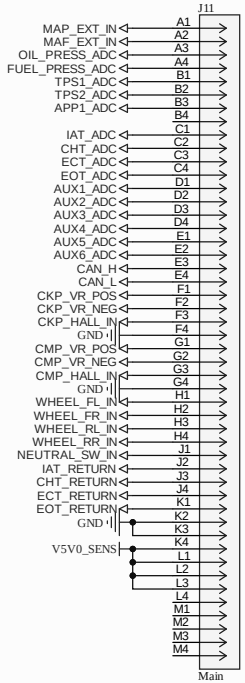
AUTOR	ATUALIZAÇÃO	REVISÃO	FORMATO
Matheus Markies	14/10/2026	A definir	A3

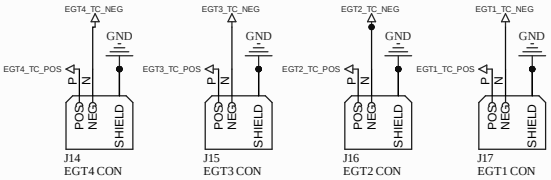
CÓDIGO DO DESENHO	ARQUIVO	FOLHA
ECTU-SCH-10.	ETC.SchDoc	10 / 12

J12 | ATUACAO / ALIMENTACAO

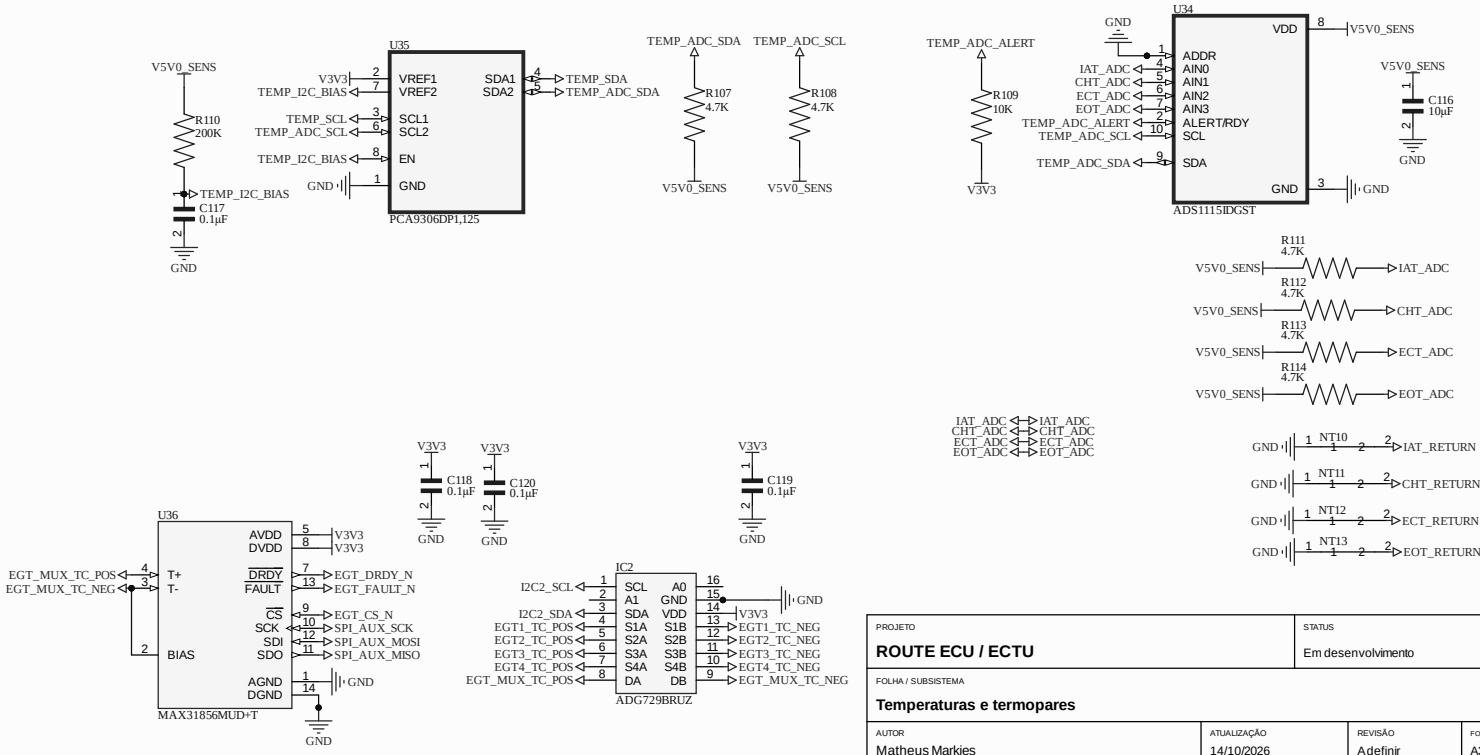


J11 | SENSORES / COMUNICACAO





PROJETO: **ROUTE ECU / ECTU**			STATUS: Em desenvolvimento
FOLHA / SUBSISTEMA: **Conectores da ECU**			
AUTOR: Matheus Markies	ATUALIZAÇÃO: 14/10/2026	REVISÃO: A definir	FORMATO: A3
CÓDIGO DO DESENHO: ECTU-SCH-11.	ARQUIVO: ECUConnectors.SchDoc		FOLHA: 11 / 12



PROJETO	**ROUTE ECU / ECTU**
STATUS	Em desenvolvimento
FOLHA / SUBSISTEMA	**Temperaturas e termopares**
AUTOR	Matheus Markies
ATUALIZAÇÃO	14/10/2026
REVISÃO	A definir
FORMATO	A3
CÓDIGO DO DESENHO	ECTU-SCH-12.
ARQUIVO	TemperatureSensors.SchDoc
FOLHA	12 / 12